

TEAM 1 · ITERATION 2

BlueJay

A Web & AI Point of Sale System

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| Project Vision

THE PROBLEM

Small grocery and retail stores rely on legacy terminal software, paper stock tracking, and manual invoice entry. Existing POS systems are expensive or desktop-only.

WHO IT'S FOR

Store administrators, stock managers, and cashiers in small grocery and retail stores.

WHAT BLUEJAY DOES

Turns a legacy console POS into a secure, multi-user web app for checkout, inventory, and reporting, with role-based access.

| Architecture Design

Backend (Spring Boot)

One package per feature (auth, product, report, user), each with its own controller, service, repository, and entities. Data flows controller → service → repository → MySQL.

Frontend (Angular)

One module per feature (product, inventory, sale, user, report), plus a core module for authentication and route guards.

Communication

REST API secured with JWT, documented in Scalar.

TARGET

Move toward Clean Architecture as the code grows, with business rules in a framework-free domain layer.

| Iteration 1 Recap

Environment

Three-container Docker setup: MySQL 8.4, Spring Boot (Java 21), and Angular

Backlog

Six epics covering every SPPP feature, with Given/When/Then acceptance criteria

Authentication

Login and logout with JWT and Spring Security, plus a 15-minute idle timeout

Process

Backend tests in CI, a two-pass system testing policy, and scope narrowed to authentication

| Iteration 2 Overview

FEATURES

- Product catalog: add products with categories, browse, and search by barcode
- User management: admins create users and view all accounts
- Core inventory report: stock on hand and latest cost, for admins and managers
- Seeded admin, manager, and cashier accounts with integration tests

PROCESS AND DEVOPS

- Estimated story points for every Iteration 2 story with planning poker
- Angular foundation: Material layout, sidebar, one module per feature
- Frontend and backend CI on develop and main
- Automatic deployment to AWS EC2 and JaCoCo code coverage

SETBACKS

- Deployment pipeline is failing due to disk space being full
- System tests are created, but are not able to test in production due to the aforementioned issue with the deployment pipeline

| Iteration 2 Testing Metrics

74

backend tests

103

frontend tests

9

system tests

| Iteration 3 Milestones

October 6–13: focus on the core POS workflow before the AI features

CHECKOUT

Execute sale, stock entry, cashier routing, and print receipt

ACCESS

Manager permissions, admin role control, and low-stock alerts for managers

REPORTING

One combined sales and inventory report with export

PIPELINE

Deployment pipeline should run for all new PRs and app should be running successfully

TESTING

All previous and current system tests should be written and ran